Sallen Key project
(CIRCUIT II)

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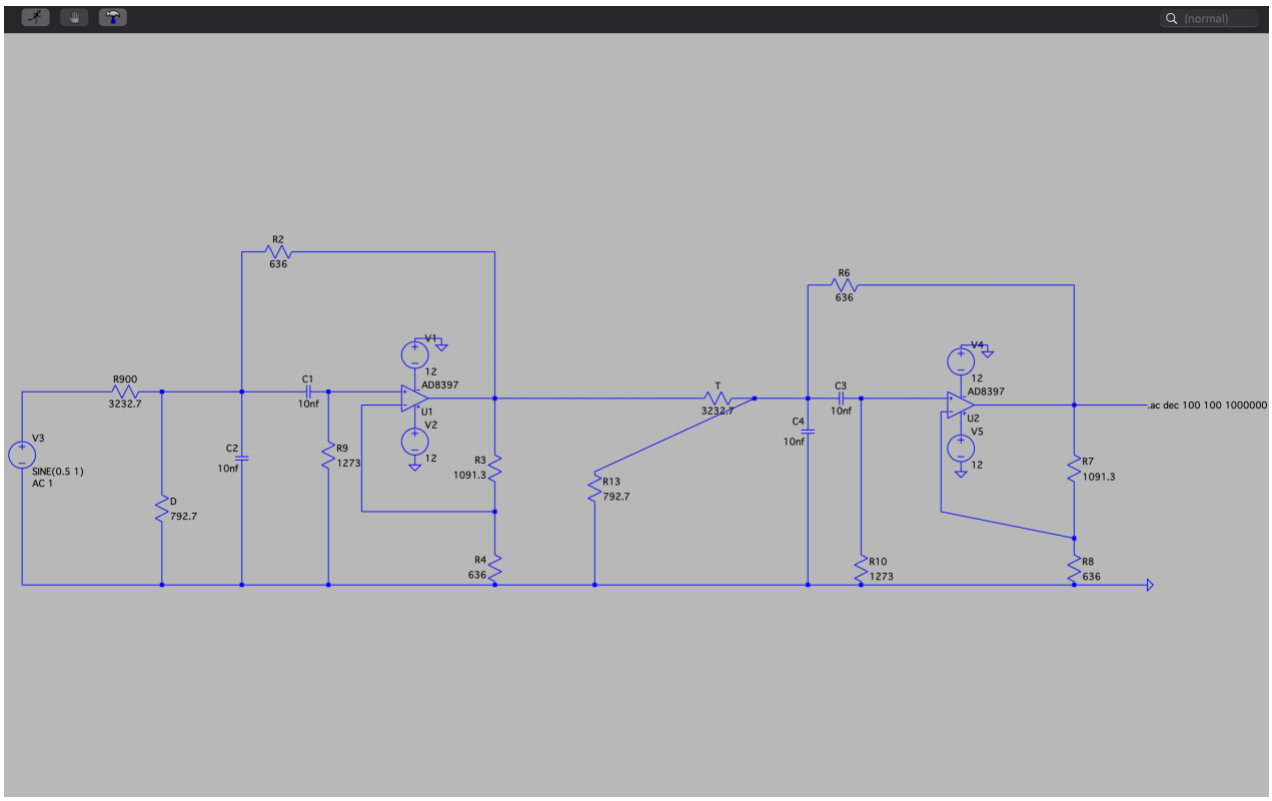
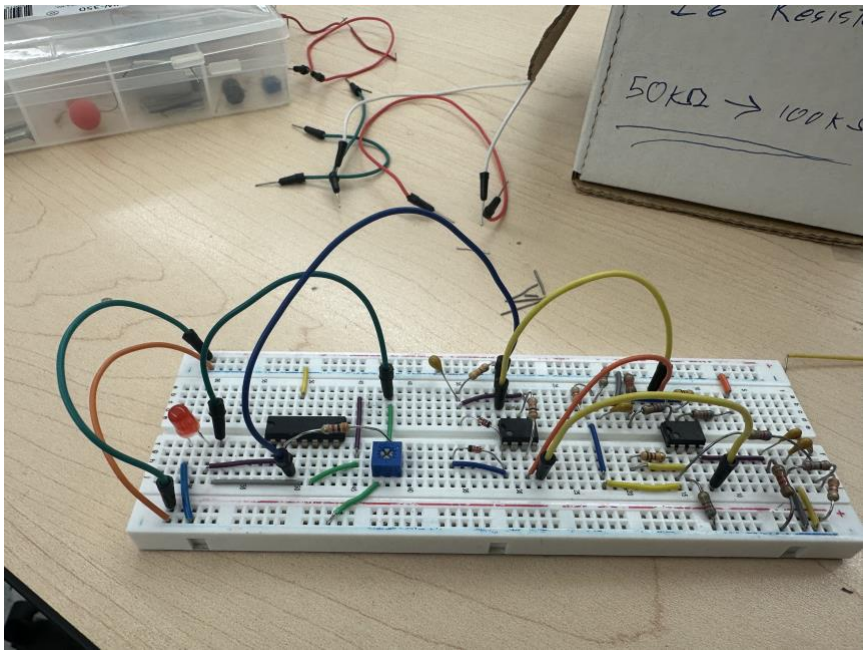
1. System Overview and description

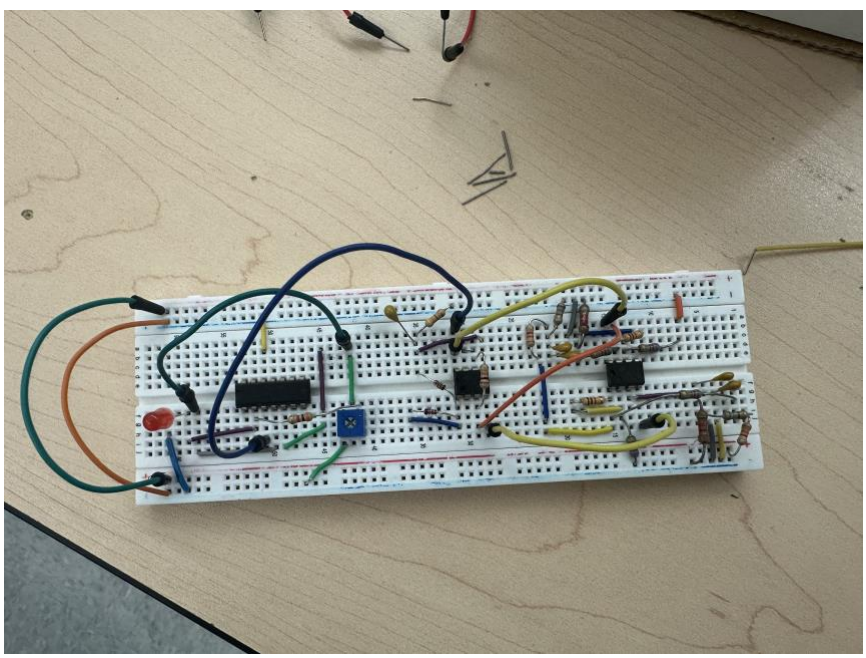
The goal of the project was to design and implement a complete IR channel detection filter system. The system built consists of a two-stage Sallen-Key bandpass filter, a peak detector, a comparator, and an LED used as an output indicator. The filter was designed to detect an infrared frequency of 25 kHz, and attenuate the unwanted frequencies of 10 kHz and 17 kHz.

LT Spice Final circuit



Fully functional physical circuit picture





2. Filter Design

Number of Stages:

The filter consists of two Sallen-Key bandpass stages, resulting in a 4th-order bandpass filter.

Q Factor:

Stage 1 Q: 3.5

Stage 2 Q: 3.5

Gain:

Stage 1 Gain: 5.35 dB

Stage 2 Gain: 5.35 dB

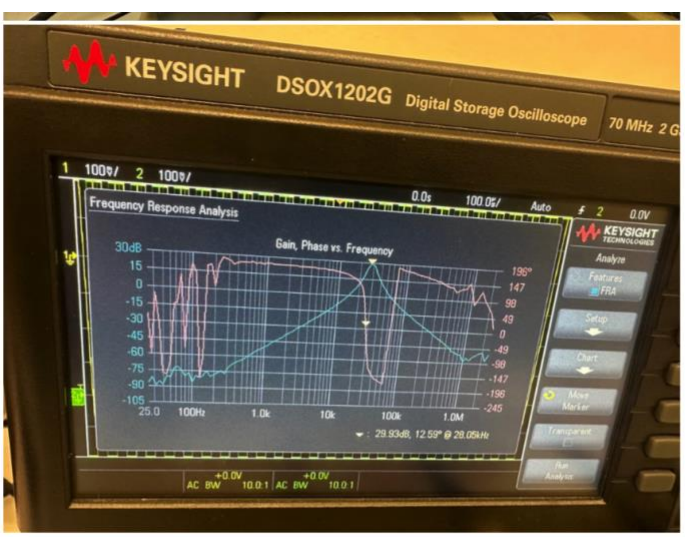
Total Gain at passband frequency: 10.70db

3. Frequency Response Verification

The frequency response was verified using MATLAB Bode analysis, LTSpice AC simulation, and physical FRA measurement.

Table			
Frequency	MATLAB	LTSpice	Measured
10 kHz	-24.08 dB	-25.39 dB	-27.90 dB
17 kHz	-8.21 dB	-10.10 dB	-13.01 dB
25 kHz	10.70 dB	10.70 dB	10.86 dB

Frequency response Analysis Result



LED Implementation and Final Results

When the 25 kHz IR channel was active, the comparator output was 5V and the LED turned ON.

When the channel was inactive, the comparator output was 0V and the LED turned OFF.

Drive link to the result video:

https://drive.google.com/file/d/1cEVZXb_HHRE72fb48Av5pNZO568tobuN/view?usp=drive_link

Conclusion

The IR detection system successfully filtered unwanted frequencies, amplified the desired 25 kHz signal, converted it to DC, compared it against a threshold, and activated an LED output.

Measured results closely matched theoretical and simulated predictions, validating the design.